

ichosynth — Usage Manual

How to play the sampler-sequencer you *draw*

You draw music on a 16×16 grid with **4 knobs** and **3 buttons**. No computer, no menus to memorize: turn, press, listen.

Level **Beginner** Build **4 encoders + 3 buttons** firmware **TCERN · SP_** See also **Build Manual**

 **You don't need a computer to play:** your **ichosynth** generates everything on its own. Plug in your **headphones**, power it via USB, and off you go.

 **This is the real TCERN.** ichosynth runs the full **TCERN firmware** (by SP_/soundpauli, toern.live) on a Teensy 4.1, built with cheap hand-soldered parts: **4 KY-040 encoders**, **3 tact switches** and an **SSD1306 OLED**. Where the original TCERN told you the state with the glowing colour of its encoder rings, our build shows it as plain text on the **OLED**.

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1 · The concept in 30 seconds

The **16×16 grid** is your music sheet. A playback "playhead" runs from left to right: every column it touches plays the notes you've put there.



- The **columns** (left→right) are the **16 steps** of one bar (a page can be chained to **32×16**).
- Each **row** is a **voice** (a sample or a synth), identified by a **color**.
- Several pages in a row make up a **pattern**; several patterns make up a **song**.

💡 Basic flow: **draw notes** → **press PLAY (B1)** → **loop**. You change samples, BPM and volume on the fly, without stopping.

2 · The hardware: 4 knobs + 3 buttons

On top there are **4 encoders** — **E1 E2 E3 E4**, left to right. Below them sit **3 tact switches** — **B1 B2 B3**. Every encoder both **turns** and **pushes** (click).

[E1] [E2] [E3] [E4] ← 4 knobs (turn + push)
[B1] [B2] [B3] ← 3 buttons

I controlli di ichosynth: 4 manopole + 3 pulsanti

Ogni manopola (encoder) si gira e si preme. Sotto, i 3 pulsanti PLAY / MENU / REC.



E1 · SINISTRA

canale / Y · nota

gira → cursore / valore

premi → mute · note shift



E2 · CENTRO-SX

pagina · nota · valore

gira → pagina / parametro

premi → conferma



E3 · CENTRO-DX

volume canale · valore

gira → parametro

premi → Play / Pausa



E4 · DESTRA

X / colonna · filtro

gira → cursore ← → · cutoff

premi → conferma / sotto-menu

B1 · PLAY

play / pausa · esci a DRAW

B2 · MENU

entra/esci menu · tieni = FX

B3 · REC

registra · tieni = count-in

L'OLED mostra lo stato in tempo reale

canale · modo · trasporto (PLAY/REC/STOP) · BPM · volume · pagina (al posto degli anelli RGB di TERN)

The gestures the firmware understands:

Gesture	What it means
Turn (E1–E4)	change the value / move the cursor in the current context
Short click (push an encoder)	confirm / act (the meaning depends on the mode)
Long press (hold an encoder)	a second action (e.g. E1 hold = mute/subpattern)
Tap a button (B1/B2/B3)	the button's main function
Hold a button	the button's held function (e.g. B3 held = recording count-in)
Combo B1+B2	a shortcut — the order you press them matters (see ch. 11 & 12)

💡 There is **no double-click**: the KY-040 encoders don't need one. Every action is a turn, a click, a hold, a button tap or a two-button combo.

What each encoder does changes with the mode. Here is the overview (details in the relevant chapters):

Mode	E1 turn	E2 turn	E3 turn	E4 turn
DRAW	Y / note (row)	page	quick channel filter	X / column
SINGLE	channel	note (pitch)	—	X / column

Mode	E1 turn	E2 turn	E3 turn	E4 turn
FILTER	slider 1	slider 2	slider 3	slider 4
MENU	—	value	value	navigate menu pages
VELOCITY	velocity	probability	channel volume	condition / timing
SONG	—	pattern	—	position (1–64)

The most useful encoder **clicks** (in DRAW/SINGLE):

- **E3 click = Play / Pause.**
- **E1 long-press** = enter mute / subpattern (release to restore).
- **E1 click @ row 16** (in SINGLE) = NOTE SHIFT.
- **E4 click** = confirm / enter a sub-menu.
- **E1 click** (in MENU) = back / exit.

3 · Reading the grid and the OLED

- **Rows** = the voices (8 sample voices + 3 synth voices), each with its own **color** (see [ch. 13](#)).
- **Columns** = the 16 steps of the current page.
- **Top row (status)**: page indicators and status lights (copy active, etc.). During Play the page indicators turn **green**.
- **Play playhead**: the highlighted column that advances as you play (the ▼ in the image above).



The original TERN showed the live state through the **colour of the encoder rings**. Our build has no RGB rings, so the **OLED** shows the same information in plain text: **current channel, mode, transport (PLAY / REC / STOP), BPM, volume and page**. Glance at it any time to know exactly where you are.

4 · The 3 buttons (B1 B2 B3)

The three tact switches are the heart of the transport and navigation:

Button	Main function	Also does...
B1 · PLAY	start playback; toggle SINGLE ; exit other modes back to DRAW	held at power-on = clear the RAM (fresh start)
B2 · MENU	enter / exit the Menu and the sub-modes	exits a sub-mode back to DRAW/SINGLE
B3 · REC	record (tap or hold)	PAUSE during playback · tap-tempo BPM · hold >300 ms = recording count-in



PLAY is B1 (a single tap), not an encoder gesture. **PAUSE is B3** while a song is running.

5 · DRAW mode (drawing)

This is the main screen, the default one, where you create patterns.

Action	Gesture
 Move the cursor	turn E1 (up/down, row/note) and E4 (left/right, column)
 Change page	turn E2
 Add / change a note	push on the cursor's spot (you hear it right away); pressing again on a note cycles the voice
 Play / Pause	E3 click (Pause during play is also B3)
 Quick filter the channel	turn E3 to soften/brighten the voice under the cursor on the fly
 Mute / subpattern	hold E1 (release to restore)

💡 E3 is the "transport" knob in DRAW: **click** it for Play/Pause, **turn** it to nudge the current channel's filter without leaving the grid.

6 · Pages, patterns and subpatterns

- The grid shows **one page** (16 steps) at a time; turn **E2** to move between pages.
- Pages with notes are played in sequence on a loop to form a **pattern**. Patterns are chained in [SONG mode](#).
- **Subpatterns: hold E1** to drop into a momentary variation of the current pattern, then release to snap back — great for live fills and breaks.
- **Note copy / paste and note-shift** let you move and duplicate steps; the active copy/note-shift state shows on the grid's top row (and on the OLED).

7 · Mute (silencing voices)

- Put the cursor on a voice and **hold E1** to mute it / drop into its subpattern; release to restore.
- Muted channels are shown **on the grid itself**, so you can always see what's silent at a glance.

8 · Volume and BPM

- **Volume** and **BPM** are always readable on the **OLED**.
- **Tap-tempo**: tap **B3 (REC)** in time to set the **BPM** by ear.
- The per-channel **volume** is adjusted in [VELOCITY mode](#) with **E3**.

9 · Velocity, probability and conditions

TCERN gives every step more than just on/off. In **VELOCITY** mode the four knobs become per-step controls:

Knob	Controls
E1	velocity (how loud the step is)
E2	probability (chance the step fires)
E3	channel volume
E4	condition / timing (e.g. play every Nth pass, micro-timing)

Use **B2 (MENU)** to step through the sub-modes; **B1 (PLAY)** returns you to DRAW.

10 · SINGLE mode (one voice only)

Useful for working in detail on one sample (e.g. a melody across several pitches).

- **Enter / exit SINGLE:** tap **B1 (PLAY)** to toggle SINGLE for the focused voice.
- In SINGLE: **E1 turn** = pick the **channel**, **E2 turn** = the **note (pitch)**, **E4 turn** = the **column (X)**.
- The same drawing/deleting gestures from DRAW apply, but only to the selected voice.

Note Shift (in SINGLE)

- Put the cursor on row 16 and **click E1** to enter **NOTE SHIFT**, then move the notes with the knobs.

11 · FILTER mode and per-voice DSP

ichosynth has the full TCERN per-voice DSP: a **lowpass filter**, **reverb**, **bitcrusher**, **detune**, **octave**, and a **Moog ladder** on the synth voices.

- **Enter FILTER mode:** press **B2 + B1 together, with B2 first** (B2-first = FILTER MODE). This works on the channels that have filters (1–8, 11, 13–14).
- In FILTER mode the four knobs become **four sliders (E1 E2 E3 E4)** for the DSP of the selected voice. The OLED shows the selected filter and its value.
- Exit with **B2 (MENU)** or **B1 (PLAY)** back to DRAW.

💡 For a fast, no-mode tweak you can also just **turn E3 in DRAW** to filter the channel under the cursor. FILTER mode is for dialling in the full set of effects per voice.

12 · Changing the sample (Sample Browser)

To assign a different WAV to one of the 8 sample voices:

1. **Open the Sample Browser:** press **B1 + B2 together, with B1 first** (or both at once). This enters SET_WAV for channels 1–8.
2. **Navigate** with the knobs: change **folder** and **sample** (browse the SD); you can preview the **length** and **waveform**, and set **seek / length / reverse**.
3. **Confirm** with **E4 click** to load the selected sample onto the voice.
4. **Exit** with **B2 (MENU)**.

📁 Samples live on the microSD as `/samples/<folder>/_<number>.wav` (see the [build manual](#)). WAVs should be mono, 16-bit, 44.1 kHz — `wavmaker.py` prepares them for you.

🕒 **Order matters in the combo: B2 first** → FILTER MODE; **B1 first (or simultaneous)** → SAMPLE BROWSER. There is a short 350 ms cooldown between combos.

13 · Voice colors and the synth voices

Each voice has a fixed color (defined in `colors.h`):

Legenda colori delle voci			
Ogni riga della griglia è una voce, identificata da un colore (da colors.h).			
	Voce 1	rosso	voce campione
	Voce 2	blu	voce campione
	Voce 3	giallo	voce campione
	Voce 4	verde	voce campione
	Voce 5	magenta	voce campione
	Voce 6	verde-lime	voce campione
	Voce 7	arancione	voce campione
	Voce 8	turchese	voce campione
	Voce 13	viola	synth (sawtooth)
	Voce 14	bianco	synth (square)

There are **8 sample voices** plus **3 synth voices**. The synth voices play internally generated waves and follow the pitches of a scale; they have their own **Moog ladder** filter (set in [FILTER mode](#)). TCERN is **polyphonic**, so several voices sound together.

14 · Sample packs

A "sample pack" is a complete set of voices saved on the SD: recall a whole kit on the fly.

- Open the Sample Browser ([ch. 12](#)) and use the pack controls to **load** a numbered pack onto the 8 sample voices.
- Packs let you swap an entire kit between songs without rebuilding it voice by voice.

 On the SD a pack is a numbered folder with its `.wav` files inside; ichosynth manages them, you don't need to create them by hand.

15 · Saving and loading (Menu)

- **Enter the Menu:** tap **B2 (MENU)**.
- Navigate the menu pages with **E4** (turn), adjust values with **E2 / E3**, and **confirm / enter a sub-menu with E4 click; E1 click = back**.
- From the Menu you **save** and **load** songs to/from the microSD.
- **Exit** the Menu with **B2 (MENU)** again, or **B1 (PLAY)** to jump straight back to DRAW.

16 · Live recording (REC)

ichosynth records audio straight from its input into the current channel.

1. Choose the **channel** to record into.
2. **Hold B3 (REC)** to record from the input (**MIC** or **LINE**; the choice shows on the OLED).
3. **Hold B3 > 300 ms** to get a **count-in** first (4 beats at the current BPM), then recording starts on the beat.
4. Release **B3** to stop; the take is saved to the SD and loaded onto the channel, so it survives a reboot.

 **B3 is multi-purpose:** a quick tap is **tap-tempo / PAUSE during play**; a hold is **record**. The OLED's transport readout shows **REC** while you're recording.

17 · SONG mode

SONG mode chains your patterns into a full arrangement.

- In SONG mode: **E2 turn** = choose the **pattern**, **E4 turn** = the **position** in the song (1–64).
 - Build the sequence of patterns, then **PLAY (B1)** runs the whole song.
 - Mute states and the song position are shown **on the grid**, so you can follow the arrangement live.
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18 · MIDI and tap-tempo

- **USB MIDI:** ichosynth is a **USB MIDI** device. Connected over USB it plays/receives MIDI with a computer or other gear.
- **Tap-tempo:** tap **B3 (REC)** in time to set the BPM by feel — no menu needed.

19 · Mode & command map

From DRAW (the main screen) you reach everything with knobs, buttons and the two combos:

Mappa delle modalità — cosa fa ogni encoder

Le 4 manopole (E1–E4) cambiano funzione in base alla modalità attiva. I 3 pulsanti restano costanti.

MODO	E1 (SX)	E2	E3	E4 (DX)
DRAW schermata principale	Y / nota	pagina	volume canale	X / colonna · filtro
SINGLE una voce a schermo	canale	nota (pitch)	—	X / colonna
FILTER combo B1+B2	slider 1	slider 2	slider 3	slider 4
MENU salva / carica	—	valore	valore	naviga pagine
VELOCITY dinamica passo	velocity	probabilità	volume canale	condizione / timing
SONG catena di pattern	—	pattern	—	posizione (1-64)

I 3 pulsanti (costanti in ogni modalità)

B1 · PLAY
play/pausa · toggle SINGLE
esci verso DRAW

B2 · MENU
entra/esci menu
tieni = FX MODE · combo = FILTER

B3 · REC
registra (tap o tieni) · pausa
tieni = count-in · tap-tempo BPM

Legend: **E1–E4** = the four knobs (turn or click) · **B1/B2/B3** = the three buttons · "click" = short press · "hold" = long press · "combo" = two buttons together (order matters).

You want to...	Gesture
Move cursor up/down (row/note)	turn E1
Move cursor left/right (column)	turn E4
Change page	turn E2
Add / change a note	push on the cursor's spot

You want to...	Gesture
Play / Pause	E3 click (Pause also B3)
Quick-filter the current channel	turn E3 (in DRAW)
Mute / subpattern (current voice)	hold E1
PLAY (transport)	B1
MENU (save/load, settings)	B2
REC / pause / tap-tempo	B3
Clear RAM (fresh start)	hold B1 at power-on
Enter/exit SINGLE	B1 (toggle)
Note Shift (in Single)	E1 click @ row 16
VELOCITY (velocity/prob/vol/condition)	B2 to reach it → E1/E2/E3/E4
FILTER mode (per-voice DSP)	B2 + B1, B2 first → sliders on E1–E4
Sample Browser (set WAV)	B1 + B2, B1 first / together → confirm E4 click
Menu confirm / enter sub-menu	E4 click · back = E1 click
Recording count-in	hold B3 > 300 ms

⚠ **The two combos differ only by order: B2 first** → FILTER mode; **B1 first (or both at once)** → Sample Browser. Wait ~350 ms between combos.

20 · Common problems

Symptom	Fix
🔇 I don't hear anything	check the volume/channel volume (OLED), that the voice isn't muted (hold E1), and that the sample exists on the SD in the right format
▶ It won't start	Play is B1 (or E3 click); Pause during play is B3
⚙ A knob goes the wrong way	it's a CLK/DT swap from the wiring stage (see the build manual)
🚫 The samples won't play	wrong SD path/structure, or WAV not mono/16-bit/44.1 kHz → use <code>wavmaker.py</code>
😬 A combo opened the wrong mode	mind the order: B2 first = FILTER, B1 first = Sample Browser; wait ~350 ms and retry
💾 I want a fresh start	hold B1 at power-on to clear the RAM; save songs to the SD from the Menu (B2)
🕒 Sample length	long samples with continuous looping are supported

Have fun! 🎵

*ichosynth runs **TCERN** by SP_ (soundpauli) · [toern.live](#) · on a Teensy 4.1 with 4 KY-040 encoders, 3 tact switches and an SSD1306 OLED.*